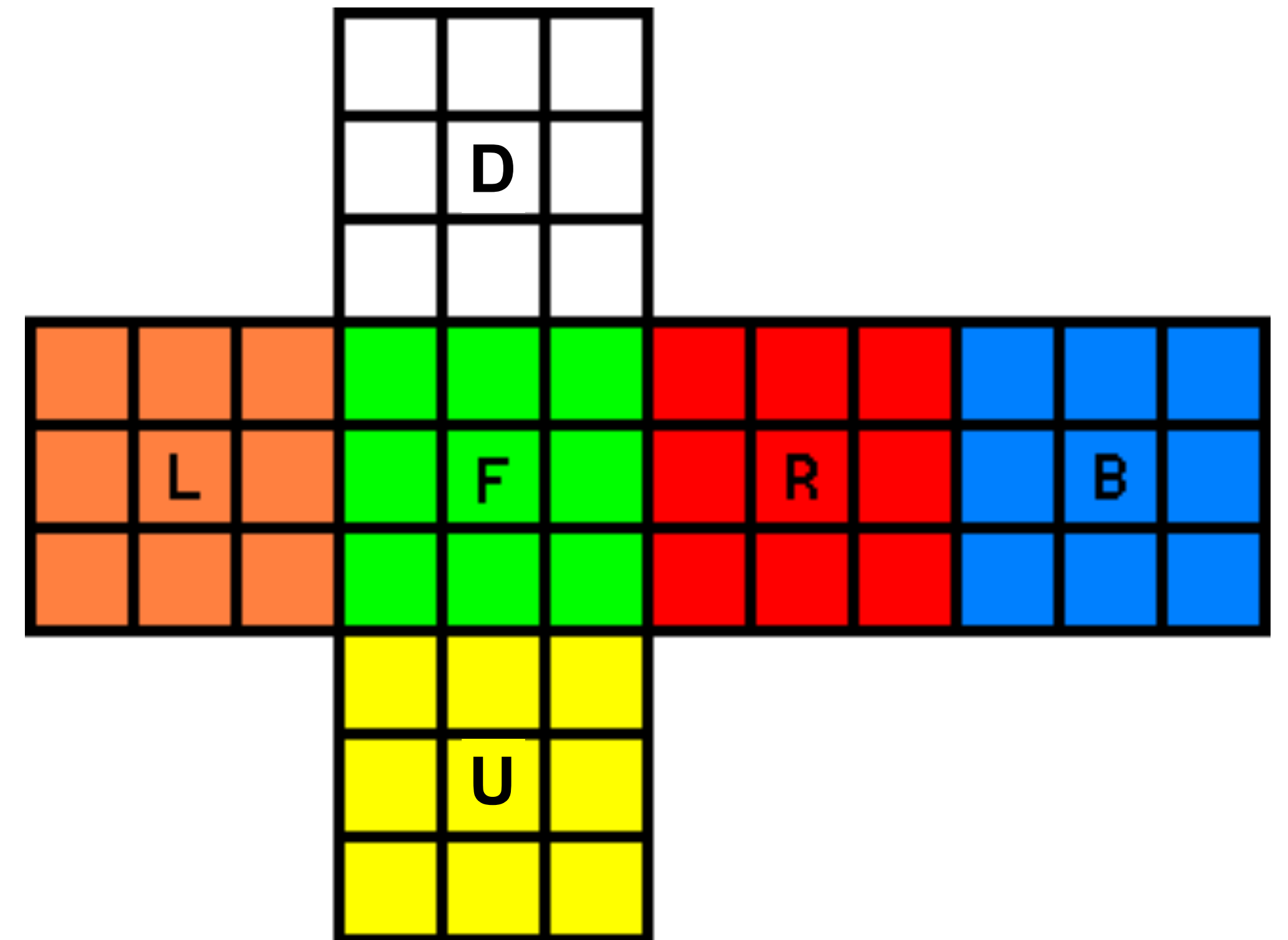
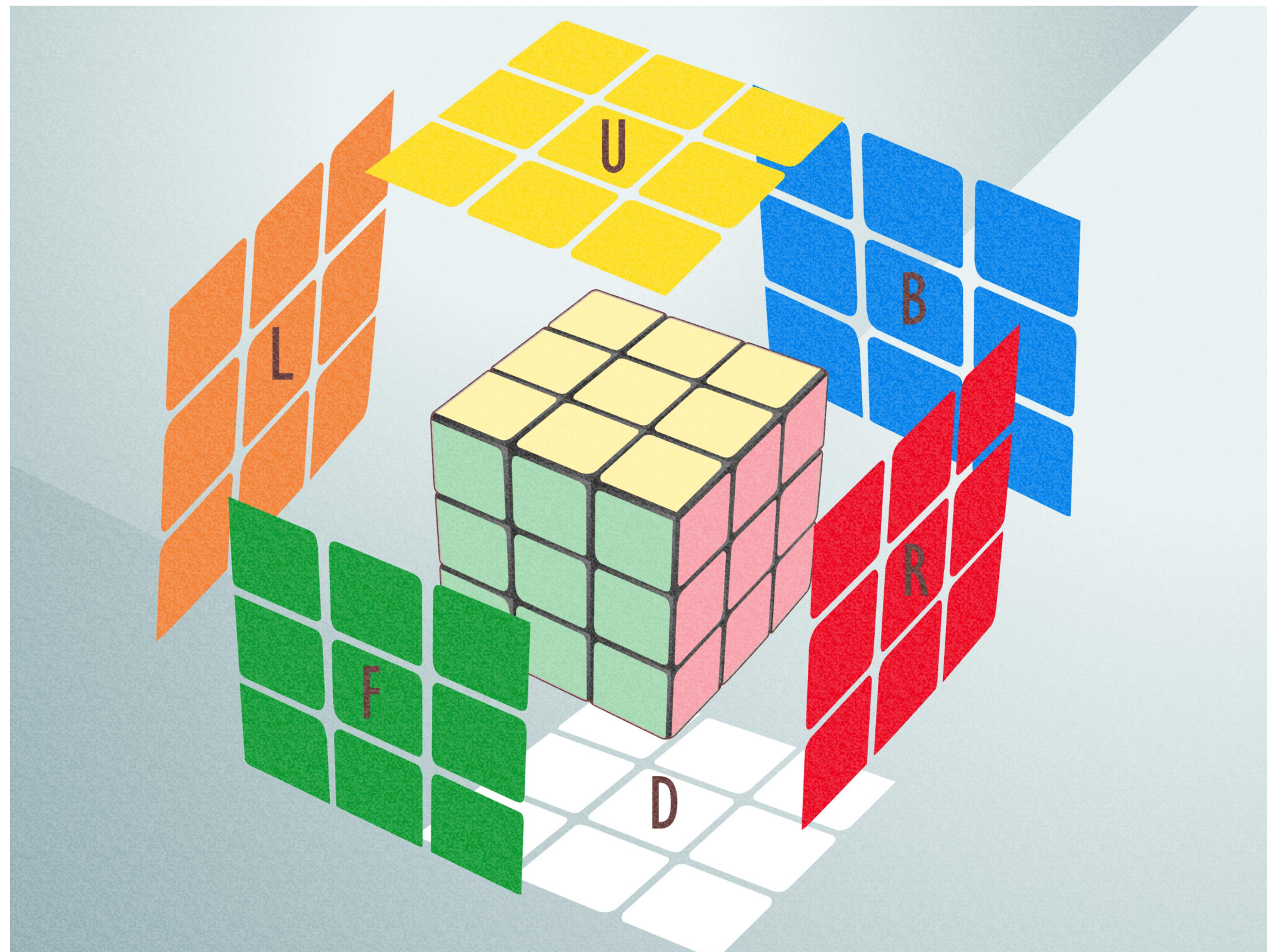


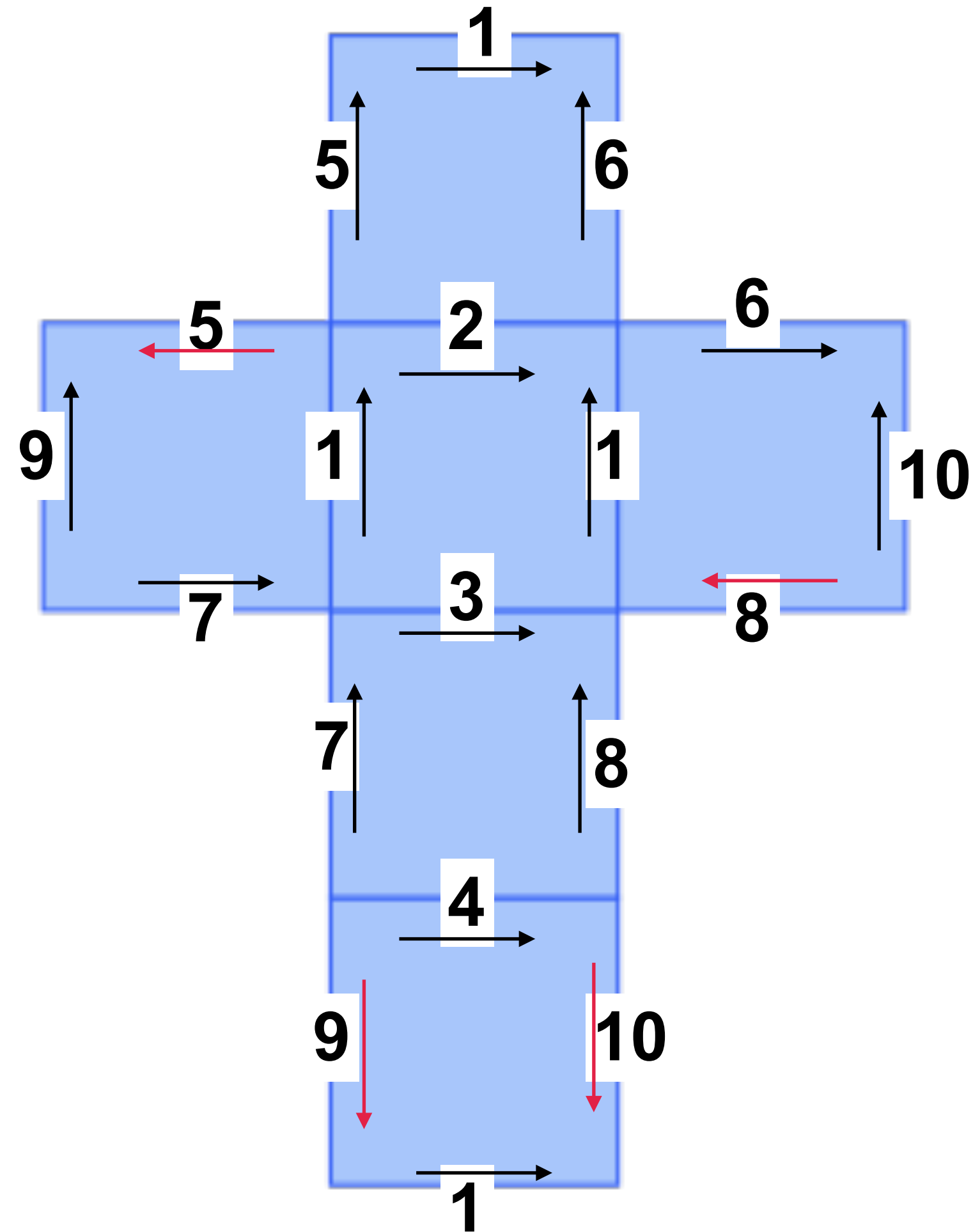
A new view for hypercubes

You can flatten a normal 3D Rubik's cube into 2D by cutting the edges where the the faces intersect like this:



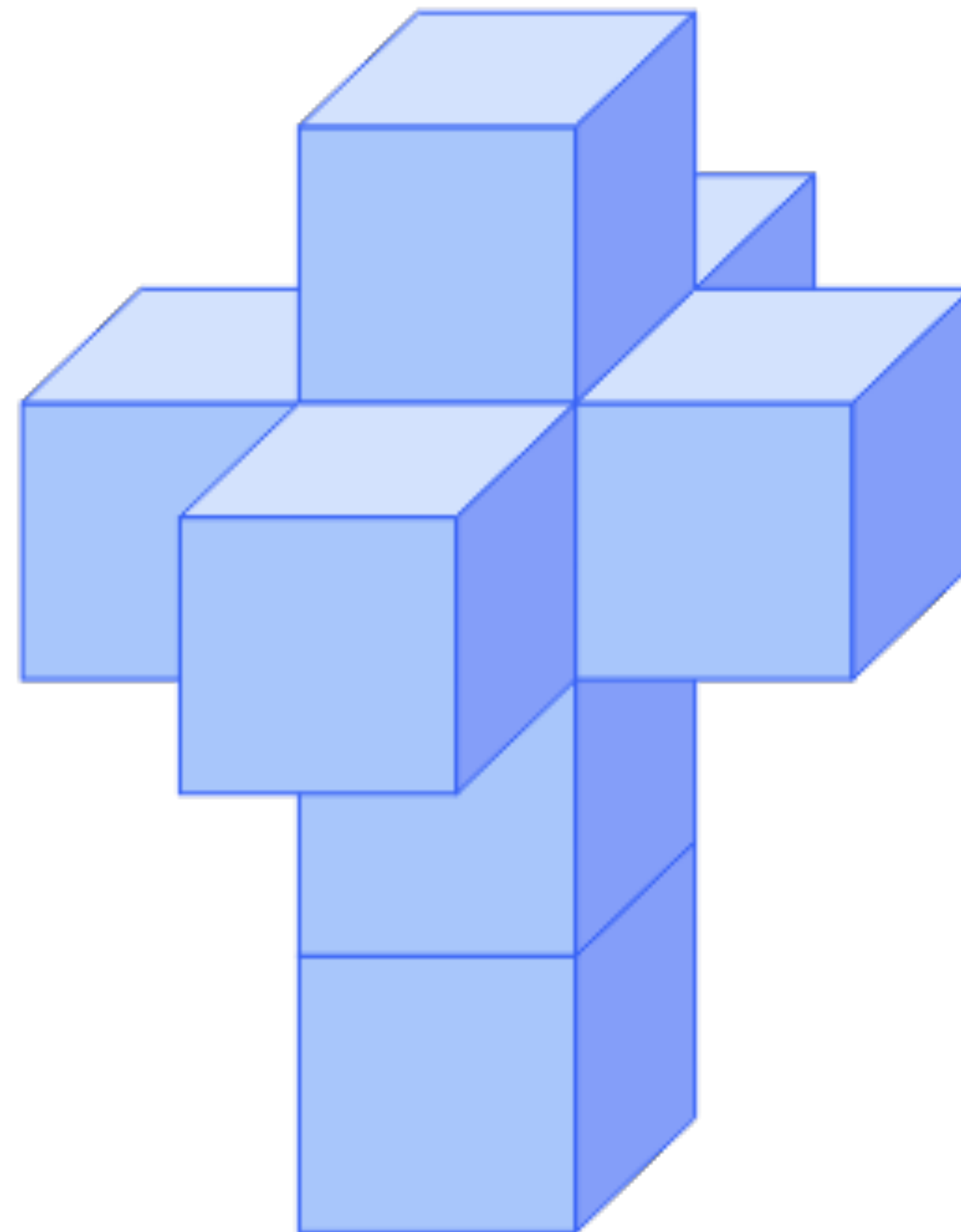
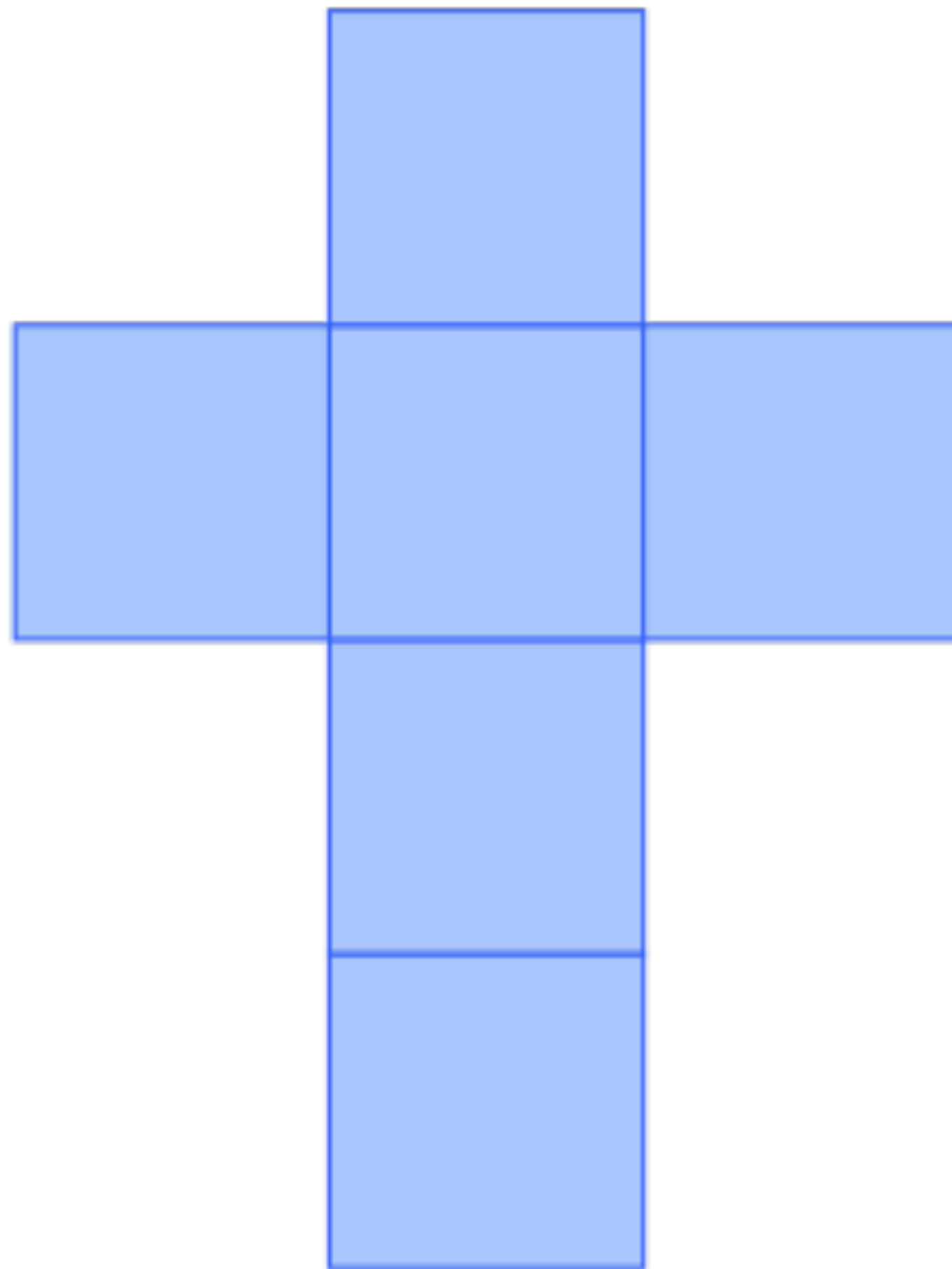
We can label each edge and keep track of orientation

Some times the orientation is reversed from the rotations along the edges needed for flattening



Just as you can flatten the 2D boundary of a 3D cube onto 2D (left)

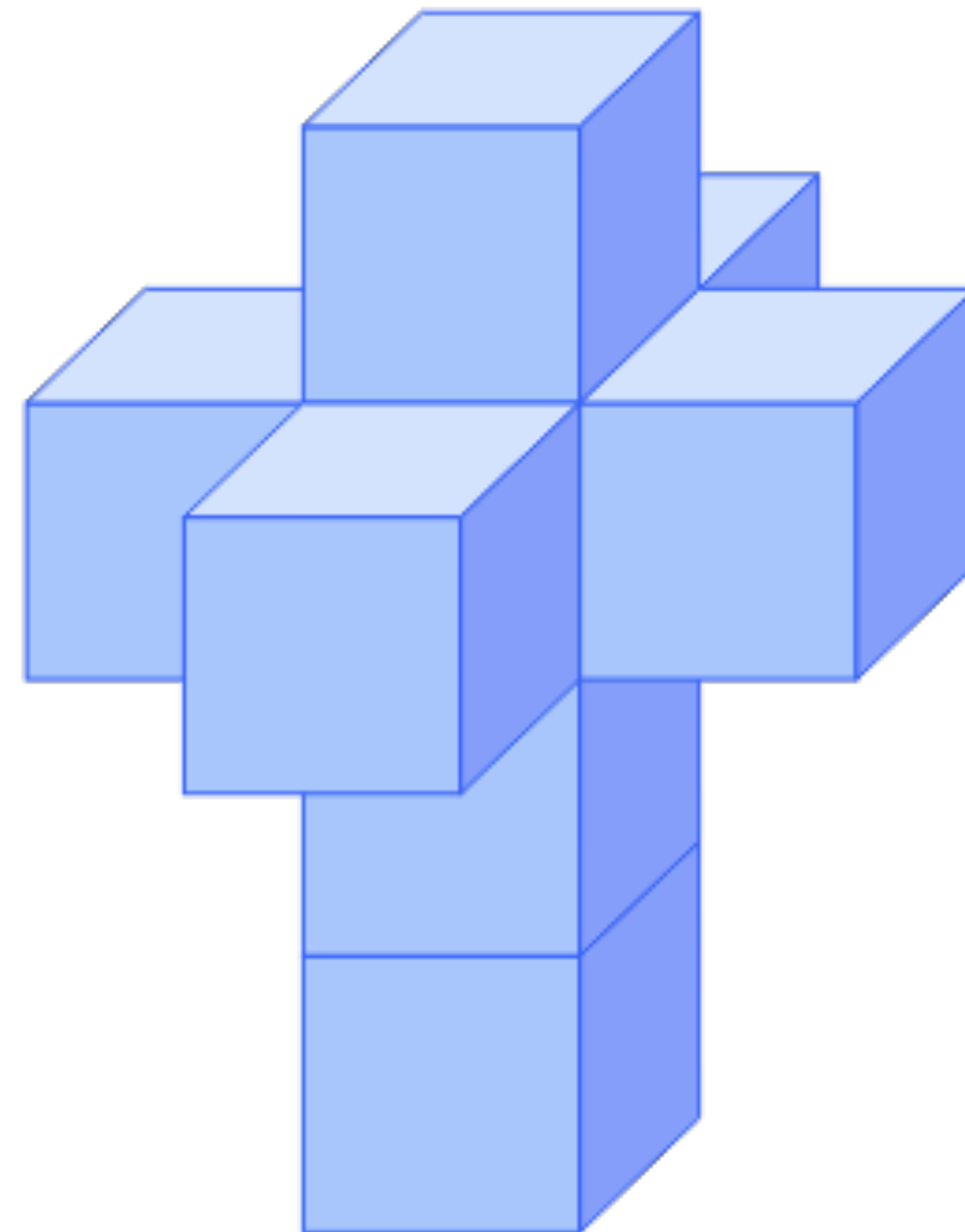
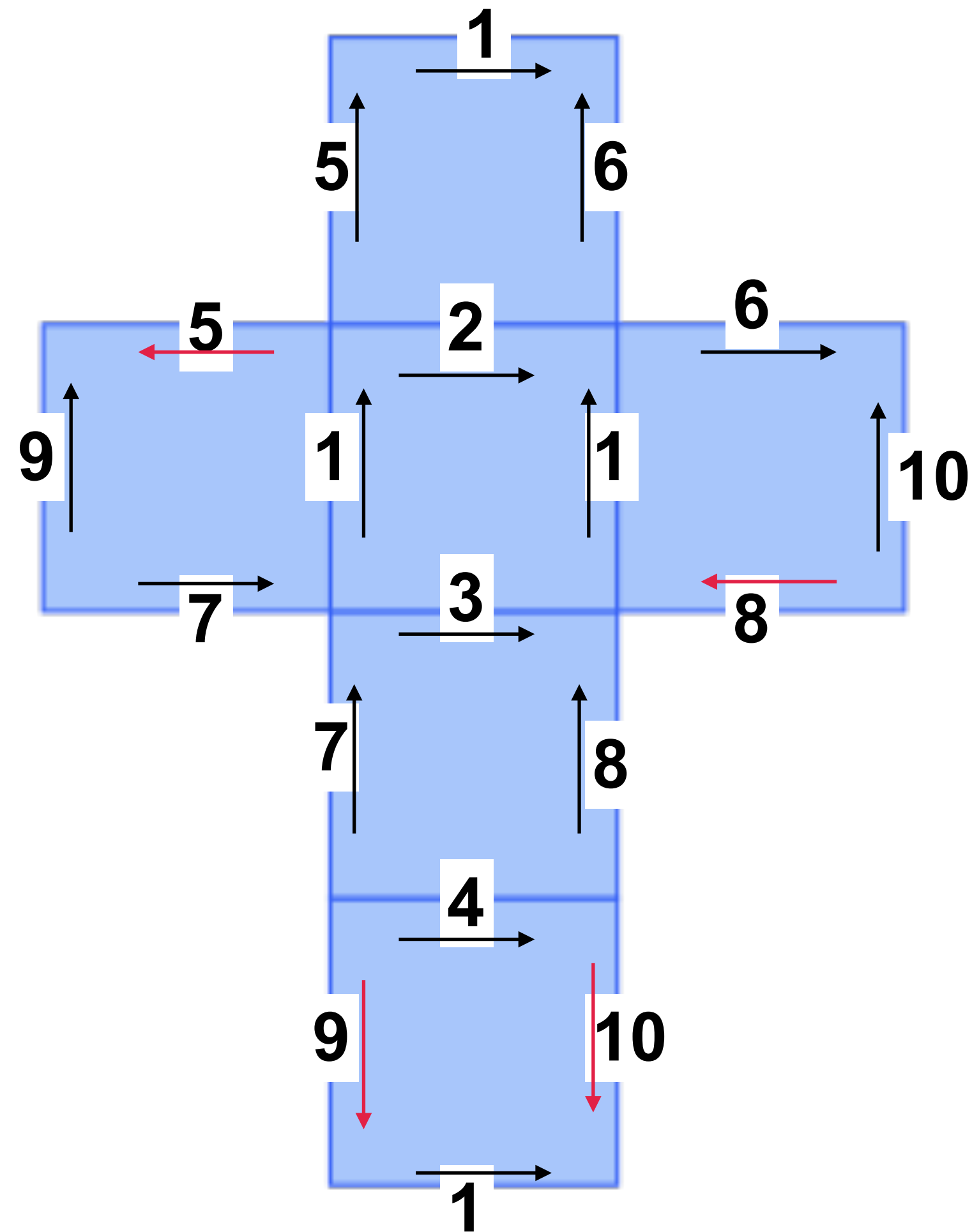
You can also flatten the 3D boundary of a 4D hypercube into 3D (right)



For the 4D cube, you can label the 2D faces that are analogous to the 1D edges for the 3D cube.

Again, we can label them, and keep track of orientation.

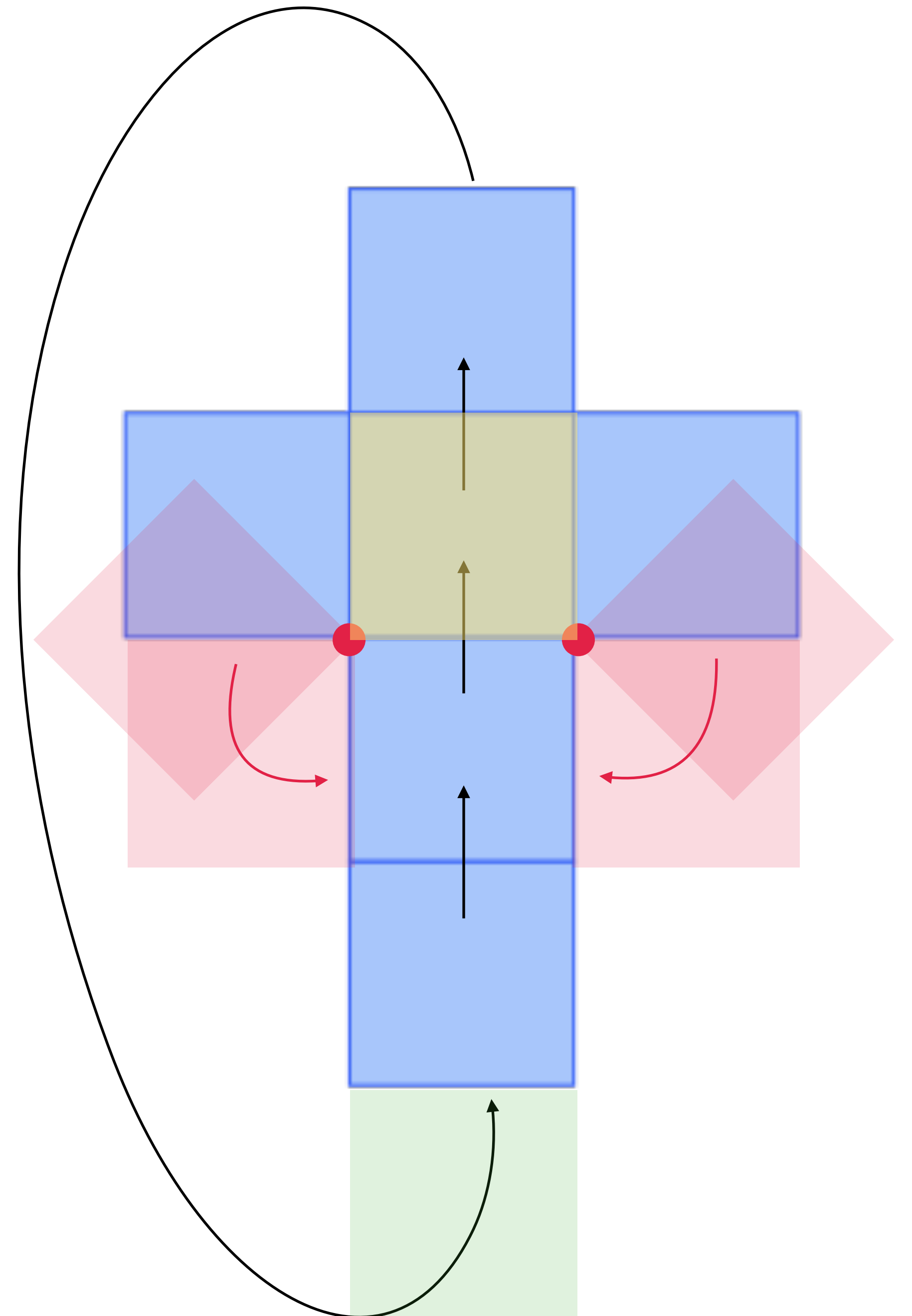
We can think of this as rotating in the 4D space so they all sit in the same 3D hyperplane.



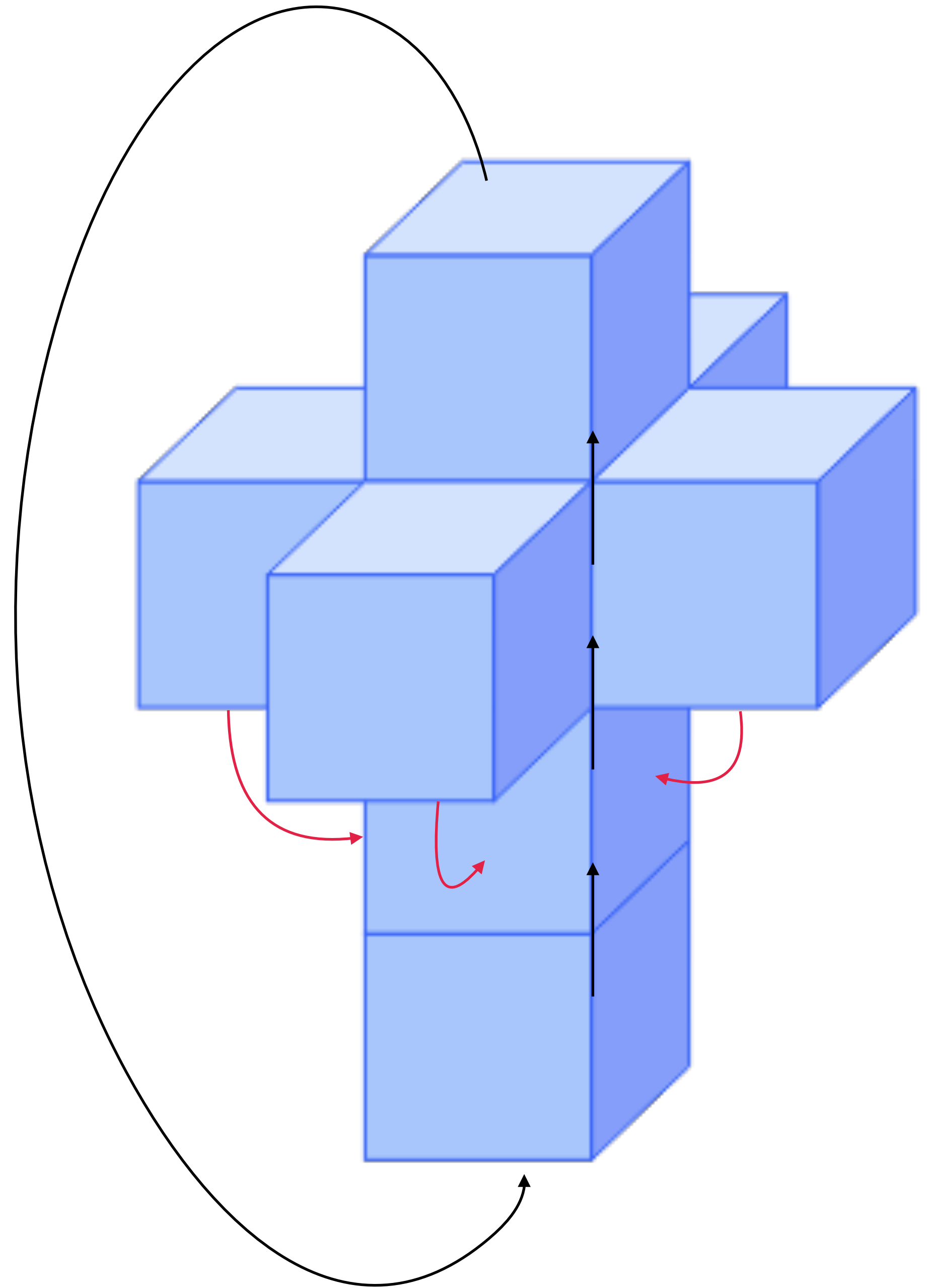
The square at the center of the cross is arbitrary.

From an interface point of it would be nice to be able to cycle.

- the center piece moves to the top
- The top piece moves to the bottom
- The two side pieces rotate

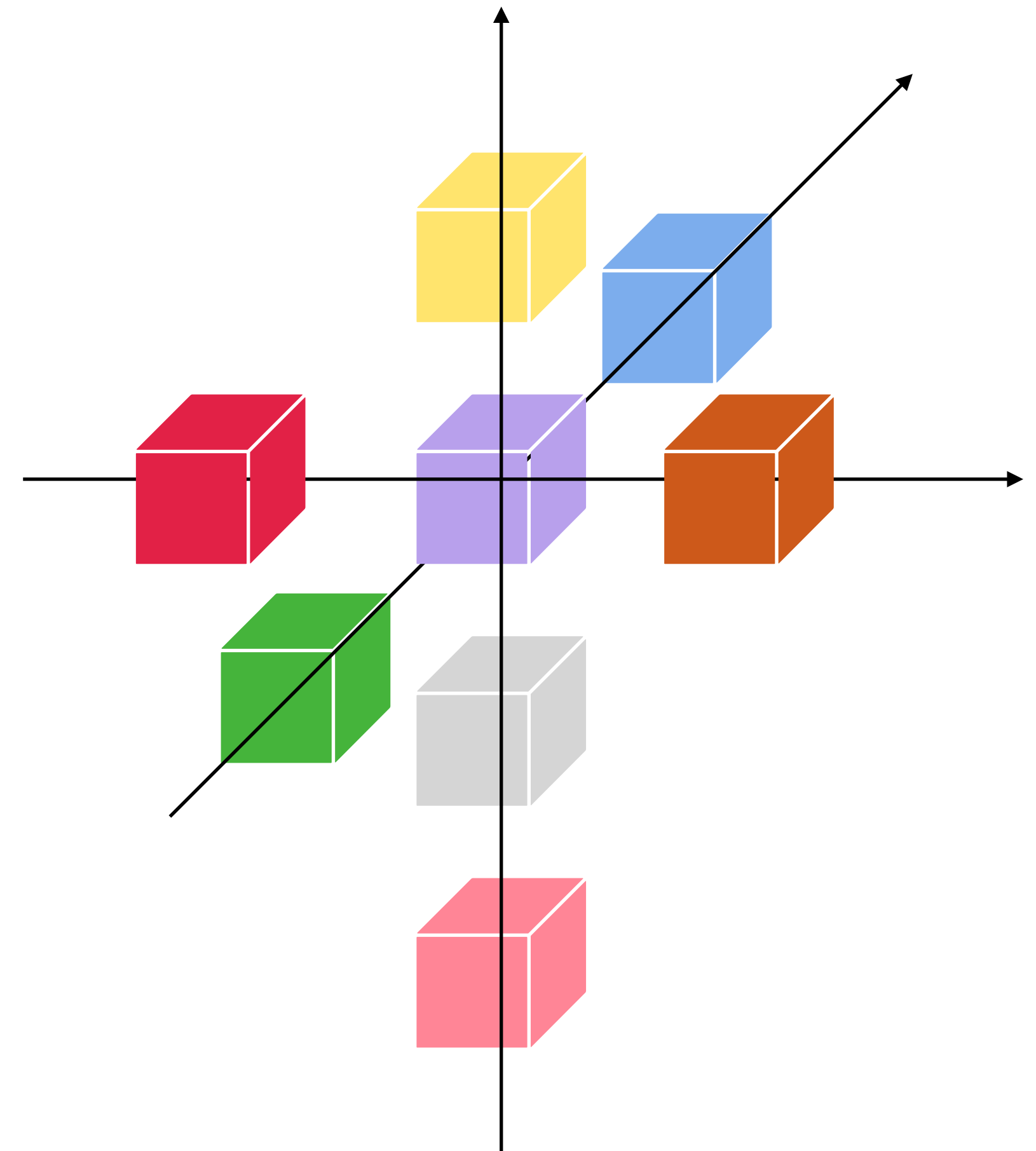


Similarly for the hypercube



For the hypercube, we will want to allow some space

So you can see the **internal central cube**



The axis that has the extra cube is also arbitrary

So we will want to be able to change that similar to the other user interface based on projections

